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Dynamic assignment of delivery order bundles to in-store customers ☆

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ABSTRACT

Many larger grocery stores offer home delivery services. However, the delivery cost is usually high and such services are rarely profitable. One way of reducing cost is by outsourcing some orders to in-store customers for a compensation. While initially single orders were dynamically assigned to customers, companies started exploring the assignment of order bundles instead to reduce per-order compensation and exploit consolidation potential. We investigate the value of dynamic assignment of bundles in this work. To this end, we consider a setting where all orders are known and, over time, unknown in-store customers enter the system for a short time and offer transportation of bundles of orders for compensation. The store decides dynamically which bundle to assign to which in-store customer (if any). At the end of the time horizon, the remaining orders are delivered by a dedicated fleet of store employees. The goal of the store is to minimize the compensation prices together with the delivery cost. We propose a threshold-based policy with scenario-based tuning. Popularity and compensation price thresholds are determined a priori by solving a set of perfect information scenarios. In every state, bundles are only assigned if they are popular enough and the compensation is comparably low. The thresholds (i.e., popularity threshold and compensation threshold) are adapted over time to account for the decrease in assignment opportunities. We show the effectiveness of our policy in a comprehensive computational study and highlight the value of bundle assignments compared to assigning individual orders. We further show that our strategy not only reduces the compensation paid to in-store customers but also the final routing cost.

1. Introduction

Many retailers such as Walmart or Home Depot offer same-day home delivery to customers. Often, customers can order until a cutoff time of day, e.g., until noon and then delivery takes place afterwards, e.g., in the afternoon. This delivery is often done by company employees [1]. As customers can be widely spread around the service area with only a few customers in some areas, the travel is often long and the cost for delivery is high. Thus, it might be valuable to outsource some parts of delivery, for example, to in-store customers that visit the corresponding areas anyway. Following [2], we call these in-store customers *occasional drivers* (ODs). Such ODs can log into the store's app and indicate their willingness to participate. They also enter the conditions under which they are willing to participate, i.e., their destination, the maximum detour they are willing to take, and their minimum compensation per bundle. Based on the information, the platform can then automatically decide to assign a bundle of orders to an OD and pay the corresponding compensation [3]. The orders in the bundle are then served by the OD and do not need to be served in the employees' tours.

Even when knowing all the ODs, the underlying optimization problem is quite rich with bundle generation, OD assignment, and final employee routing. However, in reality, ODs appear spontaneously over time. Thus, the retailer has to decide repeatedly what bundles to assign to newly arrived ODs without knowing about future ODs and their preferences. The consideration of bundles further increases the complexity of decision making and, as the recent survey by [4] summarizes, has not been studied in a stochastic dynamic setting in the literature yet. [4] further state that "Creating the right bundles at the right time and offering them to the right couriers [ODs] is extremely challenging". In contrast to the assignment of individual orders, bundle assignments are interdependent. Assigning one bundle to an OD may prohibit assigning another bundle to a second OD since both bundles may share orders. Further, a comparably cheap assignment now may break up other attractive bundles and therefore reduces the likelihood

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of the remaining orders to be assigned to future ODs. In all these considerations, the final routing cost of the company's own fleet must be considered as, for example, a comparably expensive assignment may reduce the routing cost substantially. All this must be done in real time before ODs leave the system again. This prohibits extensive calculations during the process.

To tackle these challenges, we propose a threshold-based policy with scenario-based tuning. Before the start of the process, a larger subset of promising bundles is generated. Our method then solves the perfect information problem for a set of OD-scenarios to identify a structure in bundle popularity and compensation (including approximations of the final routing of the company's employees). Based on the observed structure, time-dependent thresholds for popularity and compensation are derived. These thresholds then guide decision making during the process.

We compare our method to eight benchmark policies with respect to business concept, problem, and methodology. We also provide a perfect information bound. Our policy outperforms the benchmarks and reaches comparably small gaps of 15%–20% to the bound. We derive the following managerial insights:

- Dynamically assigning bundles of orders to ODs can reduce delivery cost significantly compared to conventional delivery by employees. In cases where the delivery cost of the company fleet is small, adding ODs is not so beneficial. Here, companies should carefully balance potential cost savings with operational simplicity.
- Only offering individual orders to ODs instead of bundles may also render delivery via ODs unsustainable. Offering carefully designed bundles instead of individual orders reduces both compensation and routing cost.
- Even assigning bundles myopically can save cost compared to conventional delivery, however, it may not only cause unnecessary high compensation cost but also breaks up bundles attractive for future ODs. The anticipatory assignment of bundles is crucial for effective decision making and yields substantially larger savings.
- For the same reason, trying to clear all orders as fast as possible may not only increase compensation paid, but also the final routing cost, as many dispersed orders must be served by the company fleet. Here, anticipating final routing cost in the assignment of bundles is very valuable.
- An effective strategy should consider the decreasing number of opportunities over the time horizon, being more selective in the beginning of the decision horizon and becoming less selective towards its end.

We contribute to the literature with respect to both problem and methodology. Our contributions are as follows. We are the first investigating the business concept where *bundles* of orders are assigned dynamically to ODs. Note that previous studies also assume that an OD can receive more than one order. However, in our work we define a bundle such that synergies between orders are explicitly taken into account by aiming at attractive combinations of requests. More precisely, we acknowledge subadditivity of costs meaning that for ODs serving a bundle of (well-selected) requests as a package, is less costly than the sum of costs for serving each request individually. This is in line with literature on bundle generation in vehicle routing (e.g., [5,6]).

Our work presents an effective anticipatory policy that allows instant decision making. We perform a comprehensive analysis to derive valuable managerial insights. With respect to methodology, To the best of our knowledge,we are among the first to use a threshold-based policy with scenario-based tuning. Both approaches work quite effectively, and, as general concepts, might prove valuable beyond the specific problem in this work.

The paper is outlined as follows. We discuss related literature in Section 2. The problem is defined and modeled in Section 3. Our solution method is presented in Section 4 followed by the computational study in Section 5. Section 6 concludes the paper with a summary and an outlook.

2. Literature

In the following, we present the related literature on dynamic crowdsourced delivery operations.

2.1. Static problems

Crowdsourced deliveries are either conducted by freelance drivers or by ODs, where the former conduct several tasks over a longer time horizon typically being active on different platforms, while the latter extend their trip only slightly [7]. Both are non-regular drivers used by companies to conduct cost efficient last mile deliveries [8,9]. Extensive literature reviews on crowdsourced deliveries including matching mechanisms [10] are provided by [4,11]. Analyses of crowdshipping adoption from the perspective of customers is presented by [12,13]. The design of crowdsourced delivery systems is investigated in [14]. In the following, we focus on ODs performing deliveries.

Typically, ODs are in-store customers with free capacity in their cars that they use to ensure deliveries once leaving the store. To the best of our knowledge, [2] were the first to present and computationally analyze the underlying decision problem, where companies include ODs into their route plans. The authors, however, assume that all ODs and their preferences are known and that each OD can cover only a single delivery. Different model formulations for this deterministic problem and the impact of different compensation schemes are analyzed in [15]. In contrast to [2], we assume that ODs are stochastic and arrive dynamically over time. We further assume that bundles of orders are assigned.

In the following years, several extensions have been proposed in the literature. For instance, [16] assign more than one delivery to each OD and also time windows are imposed. However, no dynamic or stochastic components are included in the problem setting. Solutions are generated by a hybrid algorithm combining a genetic algorithm with a local search heuristic. [17] use greedy randomized adaptive search procedures and [18] use deep reinforcement learning for crowdsourced urban delivery. Both studies assume that ODs can serve bundles of customers, but stochasticity is not considered. [19] allow for transshipment points in a crowdsourced delivery system. Again, the problem is assumed to be deterministic and static and only single orders can be assigned to ODs.

The number and location of transshipment points in this context is analyzed by [20]. The proposed model is based on static deterministic information. The reduction of fuel consumption and resulting emissions, particularly related drivers' behavior, is in the focus of [21]. ODs are assumed to serve more than one order on their routes, but again no dynamics or stochastics are considered.

The assignment of ODs to pickups and deliveries is studied by [22]. The authors assume that drivers differ in their flexibility to accept detours, but they are assumed to serve only one single order. The authors formulate the problem as a static and deterministic mixed integer linear programming model and develop a simulated annealing heuristic to generate solutions.

2.2. Dynamic problems

The dynamic version of the problem is, for instance, tackled by [23]. They assume a service platform that assigns delivery tasks to so called ad hoc drivers and a dedicated fleet. The authors propose a rolling horizon framework and an exact solution approach for the matching problem. Considerable cost reductions of about 37% compared to traditional delivery systems are shown. A dynamic and stochastic variant in which online orders as well as ODs arrive throughout the day is addressed by [24]. The authors propose a dispatching approach that

takes probabilistic information into account as well as a myopic one. A deep reinforcement learning approach for the dynamic stochastic problem is developed in [25]. In none of these studies, however, ODs have the possibility to indicate preferences (i.e., bids) on offered bundles of orders.

The multi-depot version of the problem is formulated by [26], who use an event-based rolling horizon solution approach with an iterative re-optimization procedure to generate solutions. A bi-level methodology for matching and routing of ODs and private fleets is presented in [27]. On the first level, the probability of ODs' rejection decisions is explicitly taken into account. [28] consider ODs' trajectory uncertainties by allowing multiple routine tours to be probabilistically associated with each worker. Managerial insights based on real-world data from Singapore are provided.

Another component of uncertainty is added by [29]. The authors consider uncertain travel times and a penalty for each missed delivery. A chance-constrained stochastic model imposing a probability on the maximum number of missing deliveries is formulated and two exact solution approaches are proposed.

[30] assume that drivers choose routes based on compensations and on other routes being offered. A two-stage stochastic model along with an exact as well as a heuristic solution approach are presented. To mitigate acceptance uncertainty in peer-to-peer transportation, [31] propose to let platforms generate menus of orders to choose from, which they do by a multiple scenario approach for the two-stage decision problem. A similar problem is considered by [32], where instead of menus, individual compensation is optimized to balance acceptance probability with cost. Acceptance might also depend on flexibility and/or willingness of drivers. [33] tackle the problem of dynamically matching ODs with incoming orders if compensation guarantees have to be considered. The authors propose a value function approximation for anticipatory decision making. Finally, [34] utilize in-store customers not for delivery but for dynamically picking orders in the store. The authors propose a scenario-based lookahead to calculate the value of assigning an order to an in-store customer.

2.3. Order bundles

All aforementioned work considers individual orders instead of bundles. [35] propose a bundle generation approach that aims at generating attractive packages of orders for drivers with different flexibility and willingness levels. They propose a combinatorial auction-based framework, where ODs bid on bundles depending on their characteristics. Also [36] develop a combinatorial auction for the problem of integrating ODs in traditional delivery systems. A real world study covering door-to-door deliveries in Oman shows the validity of the proposed method as well as considerable cost savings of up to 30%. [37] consider a similar problem where dynamically arriving ODs bid on individual orders and can be assigned one or more orders for delivery. They propose an auction-based multi-agent simulation to analyze the value of delivery via ODs. The algorithm aims on clearing as many orders as possible as cheap as possible in every state, thus, ignoring future developments. In our computational study, we design the benchmark-policy "Clearance" accordingly.

In all the studies mentioned above, we observe that either (i) only single orders are assigned and/or carriers do not have the option to reveal preferences (bids) for offered orders, or (ii) the problem is assumed to be static or deterministic. To the best of our knowledge, we are the first to address the dynamic stochastic problem, where bundles of orders are assigned to ODs based on their submitted bids. In Table 1, we categorize the related literature, indicating problem characteristics (dynamic, stochastic) and whether a bidding mechanism and/or anticipatory planning are included. The chronological summary shows that initial work assumed deterministic settings to focus on the theoretical value of ODs for delivery operations. Later, stochastic and dynamic aspects were added with an increasing focus on anticipation of OD appearances and their behavior. As the table also reveals, studies on the assignments of bundles to ODs is very limited and our work is the first that considers the creation and dynamic assignment of bundles to stochastically appearing ODs, where we propose an anticipatory solution method for decision making.

Table 1
Chronological overview of problem characteristics addressed in the literature.

Reference	Stochastic	Dynamic	Bundling	Anticipation
Archetti et al. (2016) [2]				
Macrina et al. (2017) [16]			X	
Cheng et al. (2017) [28]	X			
Gdowska et al. (2018) [27]	X			X
Al Hla et al. (2019) [21]				
Arslan et al. (2019) [23]	X	X		
Dahle et al. (2019) [15]				
Dayarian and Savelsbergh (2020) [24]	X	X		X
Macrina et al. (2020) [19]				
Ahamed et al. (2021) [18]	X	X		X
Triki (2021) [36]			X	
Ausseil et al. (2022) [31]	X	X		X
Dayarian and Pazour (2022) [34]	X	X		X
Di Puglia Pugliese et al. (2022) [17]				
Mancini and Gansterer (2022) [35]			X	
Shen et al. (2022) [37]	X	X		
Torres et al. (2022) [30]	X			X
Voigt and Kuhn (2022) [20]				
Alnaggar et al. (2023) [33]	X	X		X
Çınar et al. (2023) [32]	X	X		X
Di Puglia Pugliese et al. (2023) [29]	X			
Silva et al. (2023) [25]	X	X		X
Tao et al. (2023) [26]	X	X		X
Yu et al. (2023) [22]				
This work	X	X	X	X

3. Problem setting

In this section, we present the problem. We give a problem description, model the problem as a sequential decision process and provide a small example for illustration.

3.1. Problem description

We assume a home delivery problem where during a capture phase customers order delivery from the company's store to their home. In a subsequent fulfillment phase, the deliveries of the orders take place. For a systematic overview on the integration of demand management with vehicle routing, interested readers are guided to [38]. Our problem focuses on the fulfillment phase, thus, all orders are known and there is now need to be delivered until the end of the day. Hence, customers can only choose the day of delivery but cannot select specific time windows. For fulfillment, the company can access two sources, ODs and a traditional fleet. The latter represents company drivers that are employed by the company at the end of the day to fulfill orders which have not been served during the day. The ODs are in-store customers, thus, they are unknown initially and appear dynamically over the course of the fulfillment phase. They indicate via app that they are generally willing to deliver bundles of orders for some compensation and provide information about their individual preferences (e.g., maximum detour acceptable) and their destination.

At equidistant time steps (e.g., every ten minutes), the company checks the bids of the ODs currently in the system. Bids are generated automatically by the app and indicate what bundles are acceptable for the ODs and what compensation price they expect per bundle. Thus, the ODs can bid on all possible bundles without going through them one-by-one manually. Based on the bids, the company assigns bundles to ODs and compensates them accordingly. Then, the ODs deliver the

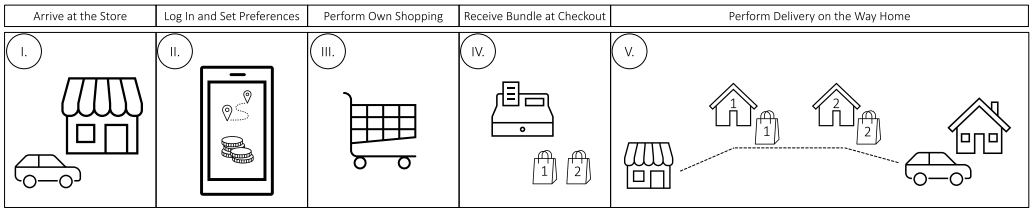


Fig. 1. Process for an in-store customer.

corresponding goods. We assume that each OD is only available in one time step and leaves the system either to deliver goods or unassigned afterwards. At the end of the fulfillment phase, the remaining orders are delivered by company drivers, e.g., company workers that agreed to deliver goods in the last hours of their shift [1]. The goal of the company is to minimize the total cost, which is the sum of the routing cost of the company drivers and the cost associated with compensation prices paid for the accepted bids.

The process from the view of an in-store customer is illustrated in Fig. 1. We assume that drivers use an app in order to facilitate the bidding process. In the first step, the customer arrives at the store. In the second step, the customer logs into the app and sets their preferences, e.g., the destination, the detour flexibility (low, normal, or high) and the expected compensation rate (low, normal, or high). Next, in step three, the customer performs their own shopping. In step four, the customer checks out and, if applicable, gets handed the assigned bundle of orders, in the example, two orders. Note that the bidding on bundles is an automated process performed by the app. Since a driver's destination and preferences are stored there, calculating the detour even for a large number of bundles can be handled efficiently. Finally, the customer leaves the store, delivers the two orders and then drives to their final destination.

3.2. Sequential decision process

We face a stochastic and dynamic decision problem as arrivals of ODs are uncertain and decisions are made repeatedly over a time horizon. Thus, we model the problem as a sequential decision process [39]. A sequential decision process describes sequences of states, decisions and corresponding cost, revelation of stochastic information, and transition to new states. A special feature in our problem is that the boundary condition differs in decisions and cost, the routing of the company's own fleet and the corresponding routing cost. In the following, we describe the components for our problem in detail. We start with the introduction of global notation.

Global notation. The set of customer orders is denoted I, where customer i requests a given quantity q_i. The depot (store, etc.) is denoted by 0. The set of nodes involved in the network is $I^0 = I \cup \{0\}$. For each pair of nodes $i, j \in I^0$, we assume a company vehicle travel cost c_{ij}^v, which is known in advance. The capacity of both a company and OD's vehicle is Q_{max}, as we assume that for urban last mile delivery the company uses small cars, which are equally sized than those of ODs. The company fleet is composed of a sufficiently large number of m identical vehicles. Each vehicle in the company fleet starts from the depot and must return to it.

States. Decisions are made in equidistant time steps within set T, where $t \in T$ and $T = \{0, 1, \dots, t_{\max}\}$. States in time t are indicated by S_t and include information about the orders still to serve $I_t \subset I$ and the n_t ODs in the system with their bids and compensation, $O_t = (B_t, C_t)$. Let $P_t = \mathcal{P}(I_t)$ be the ordered power set of I_t. Then, B_t is a binary matrix of size $n_t \times |P_t|$ with $B_{o,p,t} = 1$ if OD o is willing to serve bundle p and 0 otherwise. Note that bundle b can be a single customer order i or a set of customer orders offered as a package. As mentioned above, we assume that bids are generated automatically by an OD's app, where ODs can enter their requirements on maximum detours and expected compensations. The compensation is modeled similarly as a $n_t \times |P_t|$-matrix C_t with positive compensation values $C_{o,p,t}$ if $B_{o,p,t} = 1$ and 0 otherwise. A state can therefore be summarized as $S_t = (I_t, n_t, O_t)$

The initial state is $S_0 = (I, 0, -)$ with all orders in the system and no ODs arrived yet.

Decisions. A decision x_t in time $t = 1, \dots, t_{\max} - 1$ assigns bundles to ODs. A decision can also be represented by a $k_t \times |P_t|$-matrix with entries $x_{o,p,t} = 1$ if bundle p is assigned to OD o. Let b_t^x be the set of selected bundles by decision x_t. A decision x_t is feasible if $x_{o,p,t} \leq B_{o,p,t}\ \forall o = 1, \dots, n_t, p = 1, \dots, |P_t|$ (no assignment of bundle p if no bid by OD o), if $\sum_{p=1,\dots,P_t} x_{o,p,t} \leq 1,\ \forall o = 1, \dots, n_t$ (each OD is assigned at most one bundle), and, if the set of assigned bundles b_t^x is disjoint, i.e, an order is not assigned to two ODs.

The cost $C(S_t, x_t)$ of a decision x_t in state S_t is the overall compensation paid:

$$C(S_t, x_t) = \sum_{o=1,\dots,n_t} \sum_{p=1,\dots,P_t} C_{o,p,t} \times x_{o,p,t}. \tag{1}$$

Post-decision states. The combination of state S_t and decision x_t leads to a post-decision state $S_t^x = (I_k^x)$ containing the remaining orders to serve $I_k^x = I_k \setminus \bigcup_{b \in b_t^x} b$.

Stochastic information and transition. The stochastic information $\omega_{t+1} = (n_{t+1}, O_{t+1})$ comprises a set of n_{t+1} new ODs with their bids O_{t+1}. The new state in $t+1$ is then $S_{t+1} = (I_t^x, n_{t+1}, O_{t+1})$.

Boundary condition. In the final state (i.e., the boundary condition is satisfied) $S_{t_{\max}} = (I_{t_{\max}-1}^x)$, the remaining orders are served by the company fleet. Thus, the decision $x_{t_{\max}}$ in $S_{t_{\max}}$ determines the solution of a vehicle routing problem with cost $C(S_{t_{\max}}, x_{t_{\max}})$ being the overall routing cost.

Decision policies and cost functions. The outcome of the approach is a policy $\pi \in \Pi$. A policy assigns a decision $x_t = X^\pi(S_t)$ to every state S_t. The overall set of policies is defined as Π. An optimal solution $\pi^* \in \Pi$ minimizes the expected cost:

$$\pi^* = \underset{\pi \in \Pi}{\operatorname{argmin}}\, \mathbb{E}\left[\sum_{t \in T} (C(S_t, X^\pi(S_t)) | S_0)\right], \tag{2}$$

starting from state S_0.

3.3. Example

In the following, we give a small example to illustrate the process. Fig. 2 shows a state in decision epoch 5. The left portion of the figure depicts the orders remaining to be served after epoch 4. The middle portion depicts the state with bids and compensation prices of the ODs. The right portion shows a potential decision. In the example, after epoch 4, six orders remain in the system (A,B,C,D,E,F). The state in the middle of Fig. 2 now depicts the arrival of two ODs and their bids. Occasional driver 1 is less flexible and less costly compared to OD2. Driver 1 bids on only one bundle, (A,E), and offers to serve orders A and E for a compensation of 3. Occasional driver 2 bids on two bundles, (B,E) and (C,D). For serving orders B and E, driver 2 requests compensation of 4 and a compensation of 5 for serving orders

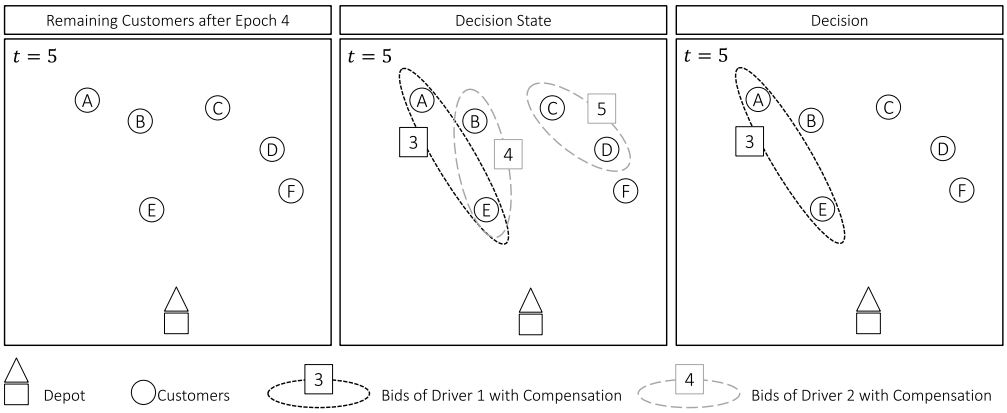


Fig. 2. Example for a state and a decision. Left: orders remaining to be served after epoch 4; Middle: state with bids and compensation prices of the ODs; Right: potential decision. Numbers in squares show bids, i.e. compensation prices.

C and D. We observe that not all orders are covered and that order E is covered by two bundles. For this example, there are five different possible decisions:

Decision	OD1	OD2	Cost	Remaining Orders
1	–	-	0	A,B,C,D,E,F
2	(A,E)	–	3	B,C,D,F
3	–	(B,E)	4	A,C,D,F
4	–	(C,D)	5	A,B,E,F
5	(A,E)	(C,D)	8	B,F

The first decision is to assign no bundle at all, leading to no cost and all orders remaining in the system. Decision 2 to 4 assign exactly one bundle and decision 5 assigns one bundle to each driver, bundle (A,E) to OD1 and bundle (C,D) to OD2. We note that decision (A,E) to OD1 and (B,E) to OD2 is infeasible as each order can maximally be assigned to one OD. Decision (B,E) and (C,D) to OD2 is infeasible as each OD can be assigned at most one bundle. The quality of a decision depends on two factors, the immediate compensation price a decision incurs and the expected cost to serve the remaining orders. Decision 1, no assignments, does not lead to any immediate cost, but all orders remain in the system with only a limited number of opportunities to assign them to future ODs. Decision 5, (A,E) to OD1 and (C,D) to OD2, leads to only two orders, B and F, remaining in the system, however, at high cost. Further, the two orders are relatively far apart making a joint service of both via one OD unlikely and a delivery by the company fleet expensive. A potentially reasonable decision, decision 2, is shown on the right side of Fig. 2. This decision assigns bundle (A,E) to OD1 at a comparably small compensation price of 3 and no bundle to OD2. The remaining four orders B,C and D,F are relatively close to each other, thus, potential future ODs may bid on the two bundles, (B,C) and (D,F). Alternatively, the routing cost of the company fleet for serving the four orders is manageable.

Even the small example already shows the challenges of decision making. In every state, an assignment needs to be found that not only is cost-efficient now but that ensures flexibility with respect to future ODs and their assignments as well as an efficient final routing of the company fleet. Furthermore, while early in the fulfillment phase, assignments can be made in a rather selective manner in anticipation of future more cost-efficient options, in later phases, clearing the number of orders in the system becomes more important to avoid high routing cost in the end. Finally, when ODs arrive, decisions need to be made in real-time. We incorporate these considerations in a heuristic solution method.

4. The bundle price consensus policy

For our setting, a method is required that assigns bundles to ODs in real-time while anticipating future bundle assignment opportunities as well as fulfillment cost of the company's own fleet. For each bundle, we determine a popularity threshold and a maximum compensation price threshold a priori. Then, in a state, we only consider bids which respect both (i.e., before we admit a bid in the auction process, we seek the *consensus* of both popularity and compensation price). These thresholds are not static, but time-dependent, and more precisely become less selective at each time-step. The idea is that, at the beginning of the time horizon, when we still have several future unknown opportunities in front of us, we can be more selective and accept only very good offers for very convenient bundles, while, as the end of the time horizon approaches and the future opportunities decrease, we become less restrictive in our bid selection in order to avoid orders remaining unassigned and requiring delivery by the company's own fleet. This procedure also avoids the issue raised in the example of myopically breaking up prominent bundles.

To determine popularity and price thresholds of bundles, a detailed and holistic view on the system is required. Our method pre-evaluates the bundles by a set of OD-arrival scenarios related to the idea presented in [40] for a dynamic dispatching problem. Each scenario is solved as a deterministic perfect information problem, in which all the ODs future arrivals are known at the beginning of the time horizon. The method then counts how often a bundle was selected and to what average compensation.

This allows identifying popular bundles that are likely effective and estimating if a bid for a bundle is cost-efficient. It also moves all the computational burden to the beginning of the fulfillment phase and allows instant decisions when ODs arrive in the system. Then, at each time step in our assignment optimization, we only consider bids addressing bundles above the popularity (bundle) threshold, and for which the price offered is below a (compensation price) threshold. Since the calculation based on perfect information values is likely too optimistic in compensation prices and does not account for decreasing opportunities over time, we adapt both the popularity and price thresholds over time, decreasing the first and increasing the second.

All the bids passing both threshold checks are admitted in the eligible bids set. Then, a static problem is solved considering only bids in the eligible sets, aiming at minimizing the compensation prices of accepted bids plus the proxy cost for orders not included in any accepted bid. The proxy cost reflects the approximate routing cost of the company fleet. If some orders are still in the system at the end of the time horizon, then a vehicle routing problem is solved to minimize the overall routing cost. We denote our policy the bundle price consensus (BPC) policy.

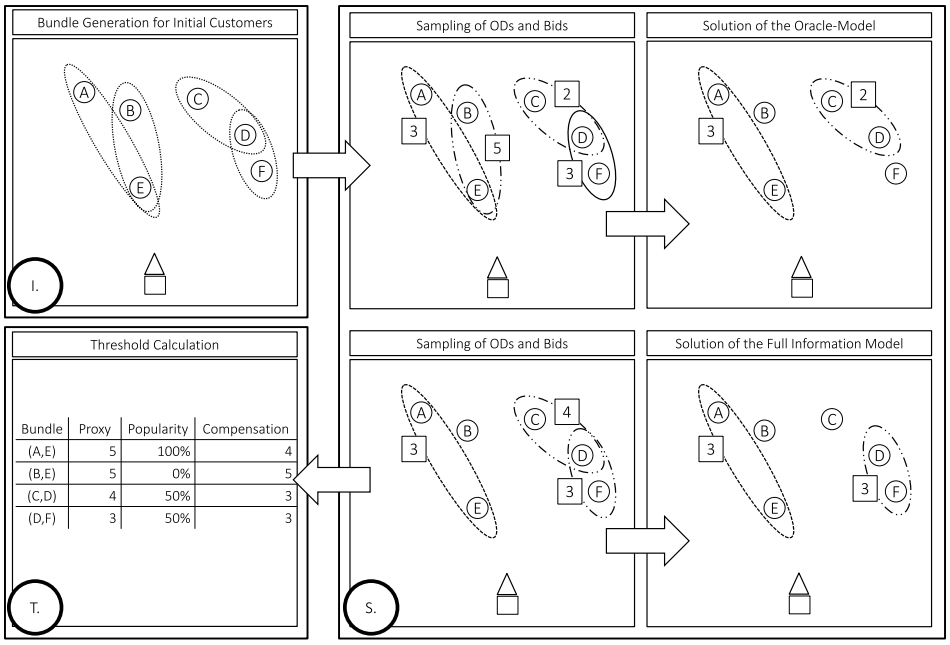


Fig. 3. Illustration of the threshold generation process.

In the framework of [39], our method can be seen as a state-dependent policy function approximation tuned a priori via scenarios. A policy function approximation represents a strategy based on practical or analytical considerations, like our thresholds. [41]. Our method is different from scenario-based methods from the literature, since we use scenarios a priory to allow for instant decision making during the process. Also, in our approach, decisions are made with respect to an adapted cost function including the proxy cost. This is known as cost function approximation. In the following, we give details on how the thresholds are generated, how the set of bundles is determined, how the routing cost per bundle is approximated and how the perfect information problem and the assignment problem at each epoch are solved.

4.1. Threshold generation

In this section, we first illustrate the process of threshold generation and then present the algorithmic details.

4.1.1. Process illustration

To illustrate the process, we rely on the example setting introduced in Section 3. The process is shown in Fig. 3. The figure contains three main parts: The initial bundle generation (I), the sampling of scenarios and their solution (S), and, the calculation of the thresholds (T). We now discuss each part for the example:

I. Bundle generation. For the set of initial orders, bundles are generated. In the example, we assume the same six orders as before. For the purpose of presentation, we assume that the bundle generation step provides four different bundles: (A,E), (B,E), (C,D), and (D,F). Evidently, the number of potential bundles is substantially higher in our experiments.

S. Scenario sampling and solution. Next, different OD-scenarios are sampled. In the example, we rely on only two sampled scenarios shown in the upper and lower right part of the figure. In the first scenario, we assume bids from three different ODs, indicated by the different lines (dotted, dashed-dotted, solid). The dotted-line OD bids 3 for bundle (A,E). The dashed-dotted line OD bids for two bundles, 5 for bundle (B,E) and 2 for bundle (C,D). The solid line OD bids 3 for bundle (D,F). The deterministic scenario is now solved by a perfect information model (see Section 4.4). The solution is shown on the top right. Bundle (A,E) and bundle (C,D) are assigned to the corresponding ODs with compensation prices of 3 and 2. Orders B and F are served by the company fleet. The second scenario on the bottom right of the figure provides different ODs with different bids. It is treated similarly and the perfect information solution is that bundles (A,E) and (D,F) are both assigned at price 3 and orders B and C are served by the company fleet.

T. Threshold calculation. In the final step, the thresholds for popularity and compensation prices are calculated. The popularity is covered by the percentage of selections over all scenarios. Bundle (A,E) was selected in all scenarios and therefore has a popularity of 100%. Bundle (B,E) was never selected, its popularity is 0%. The other two bundles, (C,D) and (D,F), were selected once leading to a popularity of 50%. The average popularity over all bundles is 50% as well. Calculating the average compensations only based on the scenarios with successful assignments will likely lead to instability and overly optimistic compensation prices. Thus, the average compensation prices are a combination of the observed compensation prices and proxy cost (each with a weight of 50%) for the routing in case the bundle was not assigned in a scenario. In the example, the proxy cost for bundles (A,E) and (B,E) are 5 (which represents the respective routing cost) since the two orders are comparably far apart. The proxy cost for (C,D) is 4. For (D,F), it is 3. For (A,E) this leads to compensation price thresholds of 4 ($100\%\times4+0\%\times5$), and for (B,E) of 5 (only proxy cost). They are 3 for both (C,D) and (D,F) as for both we weigh proxy cost and compensation prices with 50%.

Going back to the example in Section 3, the derived thresholds will likely induce the selected decision of only assigning bundle (A,E) at price of 3 and not assigning bundle (B,E) for price of 4 or bundle (C,D) for price of 5. Bundle (A,E) is very popular (popularity of 100%) and the price of 3 is below the average compensation price. Bundle (B,E) may have a smaller compensation price than the average compensation of 5, however, it is very unpopular with respect to the scenario solutions. Bundle (C,D) may have average popularity, however the compensation price is higher than the average compensation price of 3. We note that this might change in later time steps as the thresholds are softened over time.

4.1.2. Threshold generation algorithm

The algorithmic procedure for generating the thresholds is shown in Algorithm 1. Inputs are the set of orders I, the pre-generated set of

Algorithm 1 Threshold generation process

Input: Orders I, Bundles $\hat{P}$, Scenarios $\Omega_1, \ldots, \Omega_K$, **Proxy Cost** κ.
Output: Popularity σ_p^B, Price σ_p^C.

for $p \in \hat{P}$ **do**
  $\sigma_p^B \leftarrow 0$
  $\sigma_p^C \leftarrow 0$
end for
for $\Omega = \Omega_1, \ldots, \Omega_K$ **do**
  Solution $X \leftarrow SolveDeterministicModel(I, \hat{P}, \Omega)$
  for $p \in \hat{P}$ **do**
    if *IfSelected(p,X)* == true **then**
      $\sigma_p^B \leftarrow \sigma_p^B + 1/K$
      $\sigma_p^C \leftarrow \sigma_p^C + Comp(p,X)$
    else
      $\sigma_p^C \leftarrow \sigma_p^C + \kappa_p$
    end if
  end for
end for
return σ_p^B, σ_p^C

bundles $\hat{P}$ and a set of K scenarios of OD-arrivals $\Omega_1, \ldots, \Omega_K$. Inputs are further the proxy cost κ_p per bundle p in case it is not selected (we will define the proxy cost later in this section). Outputs are the average popularity σ_p^B and the average compensation price σ_p^C for each bundle p. The algorithm iterates over the scenarios and solves the deterministic model with function Solve Deterministic Model$(I, \hat{P}, \Omega)$, which is a deterministic single stage model in which ODs' bids are known in advance. The goal of this problem is to minimize the cost for accepted bids plus the proxy cost for unserved customers. The obtained solution X is then used to update popularity and price over functions $IfSelected(p, X)$, indicating if the bundle p was part of the solution and $Comp(p, X)$ the corresponding compensation. Else, the proxy cost κ_p is used.

Once we have obtained σ_p^B and σ_p^C for all the bundles in $\hat{P}$, we compute the average popularity μ^B as the ratio between the sum of popularity of all the bundles and the number of bundles:

$$\mu^B = \frac{\sum_{p \in \hat{P}} \sigma_p^B}{|\hat{P}|}.$$

Whereas, the average (compensation or proxy) price per bundle, μ_p^C is computed as the sum of the price of the bundle over all the scenarios, divided by the number of scenarios:

$$\mu_p^C = \frac{\sigma_p^C}{K}.$$

At each state t, we consider only bids addressing bundles whose popularity is higher than $\mu^B - \epsilon^B(t-1)$, and with a price lower than $\mu_p^C + \epsilon^C(t-1)$. These parameters are used to control for deviations from mean popularity and prices. This linear change allows the threshold restrictions to become less tight as the states go by, allowing the system to be more restrictive at the beginning of the time horizon, in which several potential alternative options are yet to come and less restrictive in the last states of the time-horizon, in which the remaining opportunities are limited.

4.2. Bundle generation

Since ODs are in-store customers willing to perform deliveries on their way back home, attractive bundles may contain not only orders close to each other but also those potentially very distant from each others but close to the direct path between the depot and ODs' destinations. Such long-shaped bundles cannot be generated by means of classical clustering algorithms, which aim at minimizing intra-cluster distances. To generate them, we exploit the corridors-based approach, introduced by [35]. In order to make the paper self-contained, we briefly report in the following, how this procedure works.

We initially generate a sector defined by the smallest angle α, for which all the orders are included. This sector is further split into n_s identical small sectors characterized by an angle of size $\frac{\alpha}{n_s}$. Each small sector is then explored separately. If the total demand of the orders included in the sector is smaller than the standard OD capacity, we generate a single bundle containing all of them. Conversely, if it exceeds the vehicle capacity, a clustering-based approach is applied to these orders, to generate feasible bundles. The corridor-based approach is iteratively repeated with different values of n_s, namely, 10, 20 and 30. All the bundles obtained so far, excluding duplicates, constitute the bundles in $\hat{P}$.

It is worth noting that the system we propose takes the set of bundles as input, and therefore, it works independently from the procedure used to generate them. However, the bundles' attractiveness plays a crucial role in the performance of the system, therefore is extremely important to offer attractive bundles. In fact, if we offer poor quality bundles, the number of bids received would be significantly lower and bids would be more expensive, resulting in a final solution in which most of the orders would be relegated to the last state (i.e. would be fulfilled by the company's fleet), with a large increment of costs. Indeed, we examined a variant of the newly proposed BPC, where we replace the corridor bundling approach of [35] by a spatial-based clustering approach also introduced in [35]. For the (later defined) instances with normal routing cost, with the spatial-based approach, solution quality is decreased by about 23% on average. Note that without the bundling process, the scenario approach would need to anticipate combined attractiveness of requests being assigned to one driver, where due to subadditivity of costs [42], interdependencies exist. To the best of our knowledge, this would be computationally intractable.

4.3. Proxy cost calculation

If a bundle has not been selected in a specific scenario, the compensation price is assumed to be equal to a proxy cost, κ_p, representing the cost of serving the orders belonging to bundle p by the company's own fleet. Such cost is not trivial to estimate, since it does not only depend on the order's location, but also on the location of the other orders that must be served by the company's own fleet. We compute this proxy cost depending on the order's location and on the number of orders located nearby, according to the following procedure. We split the service area in identically sized non-overlapping squares. For each square s, we refer to the number of orders located in s as N_s. We also identify primary and secondary neighborhood squares with respect to s, depicted in Fig. 4 in light blue and turquoise, respectively. The number of orders located in primary and secondary neighborhood squares are identified as $N_i'^s$ and $N_i''^s$, respectively. The distance between the depot and the center of the square s is indicated as d_s. The proxy cost for a single order i is then determined as:

$$\kappa_i = \frac{2 \times d_{s_i} \times \gamma}{N_i^s + \alpha' N_i'^s + \alpha'' N_i''^s}$$

where $0 \leq \alpha'' \leq \alpha' \leq 1$.

Then κ_p can be obtained as the average of κ_i among all orders i belonging to bundle p, divided by the cardinality of bundle p, i.e. the number of orders belonging to it. Hence, κ_p can be equal, higher, or lower than κ_i. Parameter γ represents the travel cost per with the owned fleet and is set to €1.5 per kilometer. After preliminary tests, we split the customer area in 16 squares (4 × 4 grid) and set $\alpha' = 0.1$ and $\alpha'' = 0.001$.

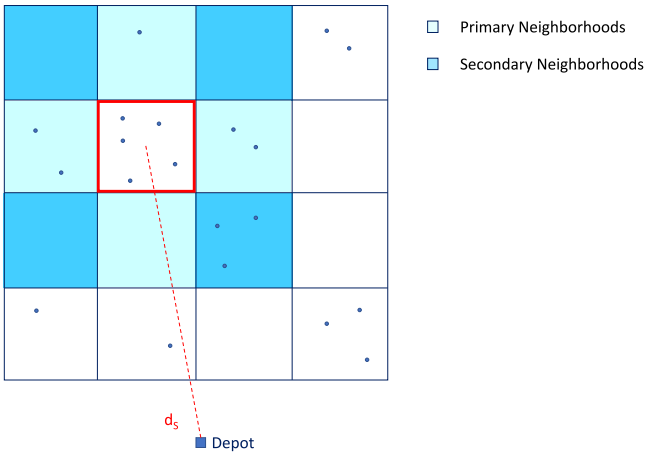


Fig. 4. Computing the proxy distance.

4.4. Perfect information model

In this section, we present a perfect information model that represents the perfect information decision problem when we know in advance the entire set of bids we receive across all the time-horizon. This perfect information model is needed for solving the scenarios. We note that for this model, we do not require any time steps and can assume that all ODs occur at once. The optimal solution of the perfect information model is a lower bound for a realization of our problem. We will use the perfect information model to solve the scenarios in the a priori part of our method, to solve the decision problem on each stage in the decision process, and as a lower bound benchmark in our experiments.

Before we present the mathematical formulation of the problem, let us introduce some supplementary notation. Recall that I^0 is the set of all the nodes involved in our network, i.e. the order locations plus the depot and c_{ij} as the cost for traveling between node i and node j. In order to make the model more compact, we define the set of submitted bids as $L = \{1, \dots, N_l\}$. For each bid l we assume to know the bundle to which it is associated, τ_l, the OD who placed it, $\hat{o}_l$, and the price offered, C_l. It is worth noting that only pairs (o, p) for which OD o is willing to serve bundle p, generate a feasible bid l. Given the fact that, generally, the ODs/bundles compatibility matrix is quite sparse, the number of variables involved in the compact formulation is much lower with respect to a more standard formulation in which variables are defined for all the (o, p) pairs.

The decision variables are reported in the following:

Z_i	binary variable that takes value 1 if order i is visited by a company-owned vehicle and takes 0 otherwise
U_{ij}	binary variable that takes value 1 if node j is visited by a company-owned vehicle just after node i and takes 0 otherwise
Y_l	binary variables that take value 1 if bid l is accepted and take 0 otherwise
Q_i	non-negative variables representing the cumulative load at node i, expressed as the total quantity of demand delivered by a vehicle along its route, when leaving node i

The model is then formulated as follows.

$$\min \sum_{i \in I^0} \sum_{j \in I^0} c_{ij} U_{ij} + \sum_{l \in L} C_l Y_l \tag{3}$$

s.t.

$$\sum_{j \in I} U_{0j} \leq m \tag{4}$$

$$Z_j + \sum_{l \in L | j \in \tau_l} Y_l = 1 \qquad \forall j \in I \tag{5}$$

$$\sum_{l \in L | \hat{o}_l = o} Y_l \leq 1 \qquad \forall o \in O \tag{6}$$

$$\sum_{i \in I} U_{ij} = Z_j \qquad \forall j \in I^0 \tag{7}$$

$$\sum_{i \in I} U_{ij} = \sum_{i \in I} U_{ji} \qquad \forall j \in I^0 \tag{8}$$

$$Q_j \geq Q_i + q_j - 2Q_{max}(1 - U_{ij}) \qquad \forall i \in I \ \ \forall j \in I \tag{9}$$

$$Q_0 = 0 \tag{10}$$

$$0 \leq Q_j \leq Q_{max} \qquad \forall j \in I \tag{11}$$

$$U_{ij} \in 0, 1 \qquad \forall i \in I^0 \ \ \forall j \in I^0 \tag{12}$$

$$Y_l \in \{0, 1\} \qquad \forall l \in L \tag{13}$$

$$Z_i \in \{0, 1\} \qquad \forall i \in I \tag{14}$$

The objective (3) aims to minimize the total cost for the company, expressed as the sum of the compensation prices of accepted bids plus the routing cost for the orders served by the owned fleet. The number of vehicles exploited by the company cannot exceed the number of available vehicles in the owned fleet, as imposed by Constraints (4). Constraints (5) and (6) ensure that each order is either served by the company or assigned to one of the ODs and that at most one bid per OD can be accepted. If an order is served by the owned fleet, it must be visited only once, as stated in Constraints (7). Constraints (8) ensure routes continuity. Constraints (9) track cumulative load at the nodes and works as sub-tour elimination constraints. The cumulative load at node 0 (the depot), is set to 0 (10). Constraints (11) ensure that vehicle capacity is respected. Finally, Constraints (12), (13), and (14) specify variable domains.

To strengthen the formulation, the following valid inequalities can be added to the model. To do that, we need to introduce a new set of variables R_{ij} that represent the load carried on arc (i, j).

$$\sum_{j \in I^0} R_{ji} - \sum_{j \in I^0} R_{ij} = q_i Z_i \qquad \forall i \in I \tag{15}$$

$$\sum_{j \in I^0} R_{j0} - \sum_{j \in I^0} R_{0j} = -\sum_{j \in I} q_j Z_j \tag{16}$$

$$R_{ij} \leq Q_{max} U_{ij} \qquad \forall i \in I^0 \ \ \forall j \in I^0 \tag{17}$$

$$R_{i0} = 0 \qquad \forall i \in I^0 \tag{18}$$

Constraints (15) state that the quantity delivered to each order exactly corresponds to its demand. Constraints (16) ensure that the total delivered quantity is equal to the sum of the demands of the orders directly served by the company. Constraints (17) ensure that the load carried by a vehicle never exceed its capacity. Finally, Constraints (18) forces vehicles to return empty to the depot. The effectiveness of exploiting these valid inequalities in terms of computational times reduction has been shown in [35].

4.5. Order assignment problem

The model used to decide on the assignment of orders to ODs at each stage, can be described as follows, where, as stated before, κ_i represents the proxy cost of order i. A new binary variable W_i is introduced to represent whether an order is fulfilled on this stage ($W_i = 0$) or postponed to the next stage $W_i = 1$.

$$\min \sum_{l \in L} C_l Y_l + \sum_{i \in I} \kappa_i W_i \tag{19}$$

s.t.

$$W_j + \sum_{l \in I | j \in \tau_l} Y_l = 1 \qquad \forall j \in I \tag{20}$$

$$\sum_{l \in L | \delta_l = o} Y_l \leq 1 \qquad \forall o \in O \tag{21}$$

$$Y_l \in \{0,1\} \qquad \forall l \in L \tag{22}$$

$$W_i \in \{0,1\} \qquad \forall i \in I \tag{23}$$

The goal is to minimize the bids acceptance cost plus the proxy cost for postponed orders, as stated in (19). The orders must be included in (exactly) one of the accepted bids or must be postponed to the next stage, as implied by constraints (20). At most one bid per OD can be accepted, this is ensured by constraints (21). Constraints (22) and (23) define the domains of the variables.

Note that, in this case, set I includes only the orders which are still in the system (i.e., the orders which are still unserved) in the current stage. Analogously, set L includes all the bids submitted in the current stage.

Once the boundary condition is satisfied, we solve a standard VRP to serve all the remaining orders with a traditional fleet.

5. Computational study

In this section, we present our computational study. We first describe the test instances and the benchmark policies. Then, we present the objective values and analyze the decision making of the different policies. Finally, we illustrate the effectiveness of our method's individual components.

5.1. Instances

The instances are derived from those introduced by [35]. They are composed of 20 orders. Because we assume private vehicles for the ODs and the employees of the company fleet, we draw the required capacity for orders from a uniform distribution in the range of 10% to 30% of the overall vehicle capacity, i.e., a vehicle can carry about five orders. We consider $t_{\max} = 10$ time steps. At each time step, the number of occasional drivers available is uniformly distributed between 1 and 6. As with the orders, the ODs' home locations are randomly distributed within the 10 km times 10 km service area. The depot is located in the south, outside of the service area in order to cover the situation of customers doing shopping on their way home.

The bids and compensation of ODs are modeled as follows. The flexibility of ODs is modeled via a maximum detour per order threshold that is randomly selected among 1, 2, and 3 kilometers. In Appendix A.3, we show that most detours for assigned bundles are actually below 1 kilometer. The ODs' willingness to participate is modeled by a parameter that represents the relative cost rate compared to average market prices. We randomly sample the parameter among 0.8, 1, and 1.2 for each OD. A value of one means that the bid is truthful with respect to the detour, 0.8 means that a 20% discount is applied, 1.2 that the bid is 20% above market price. This is based on the assumption that an OD with a high willingness to take over orders, accepts higher detours than an OD with low willingness. For each bundle, we then compute the detour required, and, if feasible, calculate the truthful bid using a travel cost (€1.5 per kilometer) for the detour plus a fixed cost per stop (€1.5) and a fixed participation cost (€2). We then multiply the truthful price for the willingness multiplier to obtain the actual bid price. Vehicle speed of the company's fleet is set to 20 km per hour. The number of vehicles is $m = 5$. This allows serving all orders with the fleet. We performed tests with different unitary routing cost for the owned fleet, namely *low* cost (€0.5 per kilometer), *normal* cost (€1), and *high* cost (€1.5).

After preliminary experiments, we set the number of scenarios to $K = 10$, the popularity threshold to $\epsilon^B = 0.2$, and the compensation threshold slope $\epsilon^C = 0.2$. In Appendix A.3, we show the impact of varying the ϵ-values. The average runtime for the a priori phase is 67.5 s. For each decision epoch, the average runtime is 5.4 s. These tests show that setting the parameters to 0 yields inferior solution quality. For ϵ^B, we see that for larger values, cost increase again, while the slope value ϵ^C is constant if larger than 0. Thus, the popularity threshold is rather sensitive to changes.

5.2. Benchmark policies

In total, we present eight benchmark policies, two problem-oriented policies, four method-oriented policies, and two business concept-oriented policies. We compare all policies to a perfect information lower bound where the perfect information instance is solved via the respective model.

The two problem-oriented policies propose alternative ways of assigning bundles to ODs:

Clearance. The goal of the policy is to clear all the orders as soon as possible to avoid high routing cost in the end. In each state, we search for the assignment that maximizes the number of orders served. In case of a tie, we select the decision which serves them at a lower cost. All the served orders are assigned to the correspondent OD, while remaining ones are kept for further time slots. To serve unassigned orders after the last time slot, a vehicle routing problem is solved similarly to our main method.

Anticipatory clearance. This policy extends the *Clearance* policy by considering potential future ODs. In each state, this policy samples future OD-arrivals for the remaining time slots. We solve the same optimization problem as in the *Clearance* policy, but we consider bids submitted by the real ODs and by the sampled ODs. All the bundles served by real ODs are assigned to them, while all those assigned to sampled ODs are kept in the system.

We further apply four method-oriented benchmark policies to analyze the value of the components of our *Bundle Price Consensus* (BPC) policy that uses two time-dependent thresholds on bundle popularity and compensation price.

Bundle consensus (BC). This policy is similar to BPC, however, instead of checking both the popularity and price threshold, this policy only considers the popularity threshold. Thus, in the example in Section 3, it will assign both bundles (A,E) and (C,D).

Price consensus (PC). This policy only considers the price-thresholds. Thus, in the example in Section 3, it will be indifferent between bundles (A,E) and (B,E).

BPC-fixed threshold (BPC-FT). This policy is similar to our BPC-policy, however, we assume that the thresholds are fixed over the time horizon. The fixed threshold is equal to the one used in BPC at the first time step. It can be obtained by fixing ϵ_B and ϵ_C equal to 0.

BPC-no threshold (BPC-NT). This policy is similar to BPC but does not set any thresholds. Thus, it solely relies on balancing immediate cost and proxy routing cost. It is therefore a cost function approximation and its performance is an indicator of the effectiveness of the proxy cost approximation.

The two business concept-oriented policies allow analyzing the value of offering bundles to ODs and using ODs for delivery:

No ODs. For this policy, we serve all orders with the company's own fleet. To this end, we solve the vehicle routing problem for all orders. We set the maximum runtime per realization to one hour.

No bundles. This policy follows exactly the procedure of BPC, but only considers bundles of size 1. It therefore allows to analyze the value of bundling orders for delivery.

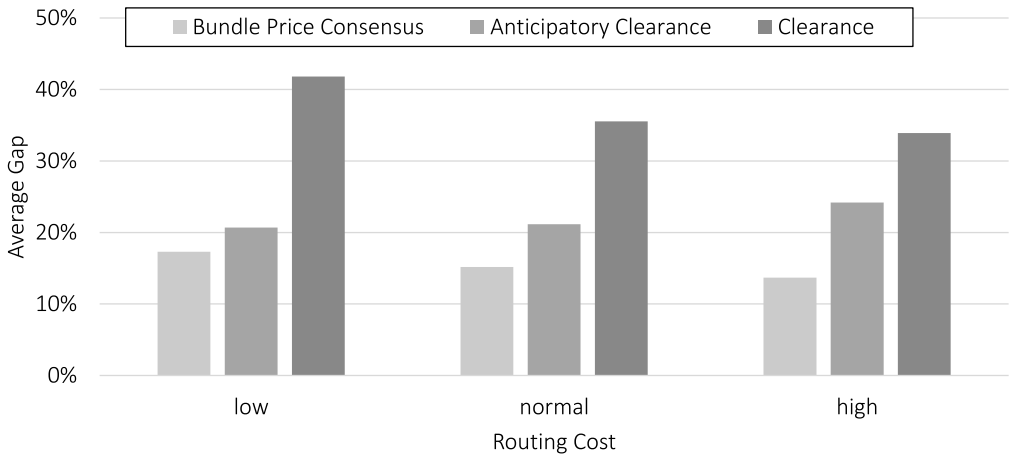


Fig. 5. Comparison of the perfect information gaps for the method-oriented benchmark policies for different routing cost.

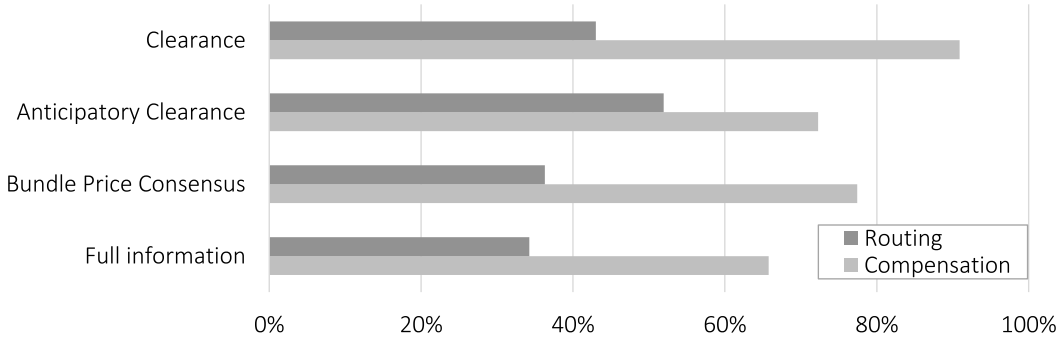


Fig. 6. Routing and OD related costs for different policies and normal routing cost.

5.3. Objective and decision making

First, we show the effectiveness of our policy compared to the problem-oriented benchmark policies and analyze the difference in their decision making. To this end, we calculate the objective values over ten instance realizations for each routing cost setting for the policies BPC, *Clearance*, and *Anticipatory Clearance*. We then calculate the average gap with respect to the perfect information solutions. The results are shown in Fig. 5.

The x-axis depicts the different routing cost settings. The y-axis shows the average gap. We observe that our policy outperforms the benchmark policies significantly. The gaps of BPC are always between 10% and 20% while the gaps for *Anticipatory Clearance* range between 20% and 30% and the gaps for *Clearance* range between 30% and 50%. The latter indicates that greedily clearing the number of orders in the system comes at high cost of compensation, and, as we show next, does not necessarily lead to lower routing cost for the company's own fleet. In our computational study, we assume instances with 20 customers to keep the computational burden of our experiments manageable. To test scalability and effectiveness of our approach, we also tested instances with 40 customers for normal routing cost. The gaps to the perfect information solution for Clearance and Anticipatory Clearance increase by 38.4% and 29.9.% on average, respectively, while BPC yields average gaps of only 12.9% . This confirms its effectiveness also for larger instances.

To analyze the cost of compensation prices and routing, we focus on the *normal* routing cost setting and calculate the average cost for compensation prices and routing for all policies and the perfect information solution. For the purpose of presentation, we normalize the cost values with respect to the cost of the total perfect information solution. The results are shown in Fig. 6. The x-axis depicts the relative cost of routing and compensation prices with respect to the total perfect information cost. That means that for the perfect information solutions, the sum of routing and compensation is 100% overall. We note that for the perfect information solution, the compensation cost is twice as high as the routing cost. We see that our BPC-policy achieves nearly the same routing cost as the perfect information model and slightly higher compensation cost. *Clearance* results in very high compensation cost while *Anticipatory Clearance* can reduce compensation cost even below BPC, however, at the expense of very high routing cost.

It is noteworthy that the routing cost for *Clearance* is higher than for BPC even though *Clearance* aims on leaving as few orders as possible in the system to save routing cost in the end. However, it ignores that assigning some bundles now may destroy profitable bundles for the future and may leave orders stranded throughout the service area. Our BPC-policy carefully evaluates potential assignments and may decide against assigning a bundle in anticipation of future opportunities and routing cost.

This is also reflected when analyzing the average number of orders in the system of the time steps. This is shown in Fig. 7. The x-axis shows the ten time steps. The y-axis indicates the average remaining number of orders after the time step. The value at time step 10 therefore indicates the average number of orders that need to be served by the company's fleet. We observe that the number of orders for the perfect information solutions are usually highest while the number of orders for the *Clearance* policy are always lowest as could be expected. The first can be explained by the fact, that the perfect information "knows" the future ODs and, therefore, can postpone assignments without risk. Interestingly, our BPC-policy can catch up with the *Clearance* policy and, in the end, has nearly the same number of orders remaining to be served by the same fleet (though, at substantially lower cost as we have shown in Fig. 6). Thus, with a few careful postponements in

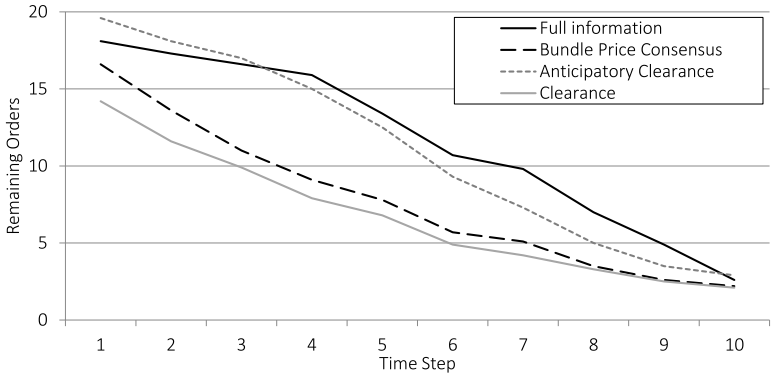


Fig. 7. Comparison of average number of orders in the system at different time steps for different policies and normal routing cost.

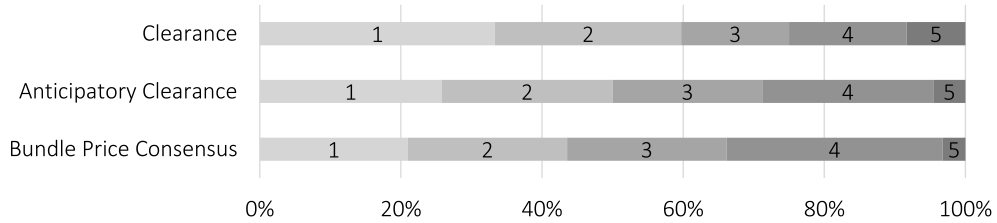


Fig. 8. Relative distribution of bundle sizes.

the beginning, BPC allows cheaper and more assignments later, likely, because prominent bundle opportunities are saved by not assigning "wrong" bundles earlier. We also note that while the average number of remaining orders are around 2 to 3 for BPC, the individual instances differ significantly. In about half the instances, about 4 or 5 orders remain, for the other half, all orders are cleared. Since vehicles have capacity for about 5 orders, our method either sends out one vehicle or no vehicle of the company's owned fleet. While we see a clear tendency in these remaining number of orders, an analysis of the average compensation and number of bundles did not show significant value differences over time and instances.

Next, we analyze the type of bundles that are assigned. To this end, we count the number of assignments for different bundle sizes for the three policies *Clearance*, *Anticipatory Clearance* and BPC and calculate the relative frequency for each size. The results are depicted in Fig. 8. The x-axis depicts the relative frequency and the number on the bars indicate the corresponding bundle size. Although the maximum size of the bundles offered is six, only very few bundles contain six orders, and none of them is selected by any policy. Therefore, we discard them in this analysis. We recall that the overall number of assigned orders are about the same for all policies as seen in Fig. 7. However, the bundle sizes are very different. Notably, the *Clearance*-policy and also the *Anticipatory Clearance*-policy assign substantially more bundles of size one but also more bundles of size five compared to BPC. Our policy shows a more balanced distribution, likely because too small bundles might prohibit assignments of prominent bundles in the future.

5.4. Method

Next, we analyze the components of our method, namely the value of the threshold-based policy with scenario-based tuning, the two time-dependent thresholds and the routing cost approximation. To this end, similar to Fig. 5 we compare the average gap of our BPC with the four method-oriented benchmarks, only the bundle popularity threshold (BC), only the compensation price threshold (PC), fixed thresholds BPC-FT, and no thresholds at all (BPC). The results are shown in Fig. 9.

We observe that BPC always provides the best solutions with gaps around 15%. The two policies with individual thresholds, BC (around 20%) and PC (20%–30%), perform comparably well but still significantly worse. This indicates that a joined consideration of popularity and price is required. Notably, BC outperforms PC for all settings. Thus, for our setting, bundle popularity seems to be more important than compensation prices. Policy BPC-FT performs significantly worse than the policies with time-dependent thresholds, reaching gaps of 30%–40%. Its performance for the setting with high routing cost is especially poor. The policy ignores that assignment opportunities decrease over time. Therefore, it ends up with more orders at the end to be served by the company's fleet. This is particularly expensive if the fleet's routing cost is high. The opposite development can be seen for the policy without thresholds BPC-NT. For low routing cost, it performs similarly to the greedy *Clearance*-policy with a gap of about 40%. With increasing routing cost, its relative performance increases as well with a gap of about 20%. As this policy balances compensation prices with approximated routing cost, it anticipates the cost of not assigning a bundle. Once the routing cost become more important, this anticipation becomes more valuable as well. The relatively strong performance of this policy also confirms the value of the routing proxy cost approximation.

5.5. Business concept

Finally, we analyze the value of the business concept of assigning bundles of orders to dynamically occurring ODs. To this end, we compare BPC to a policy only assigning single orders to ODs and a policy not relying on ODs at all. The results are prepared similarly to the other experiments. They are presented in Fig. 10. We note that the range of the y-axis is significantly larger than for the previous policies as some experiments result in cost more than twice as high compared to the perfect information solution.

First, we observe that the No Bundles-policy is always substantially worse than BPC. This indicates that it is very valuable for a company to offer bundles instead of individual orders. The reasons for the gap are twofold. First, when only offering single orders, the cost per order as a combination of fixed cost and detour cost is higher compared to a case where orders are bundled. The participation cost is only paid once and

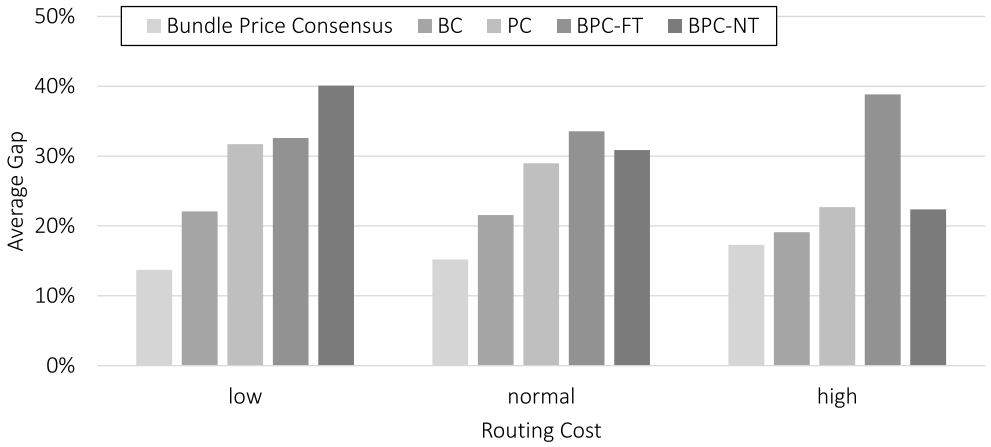


Fig. 9. Comparison of the perfect information gaps for the method-oriented benchmark policies for different routing cost.

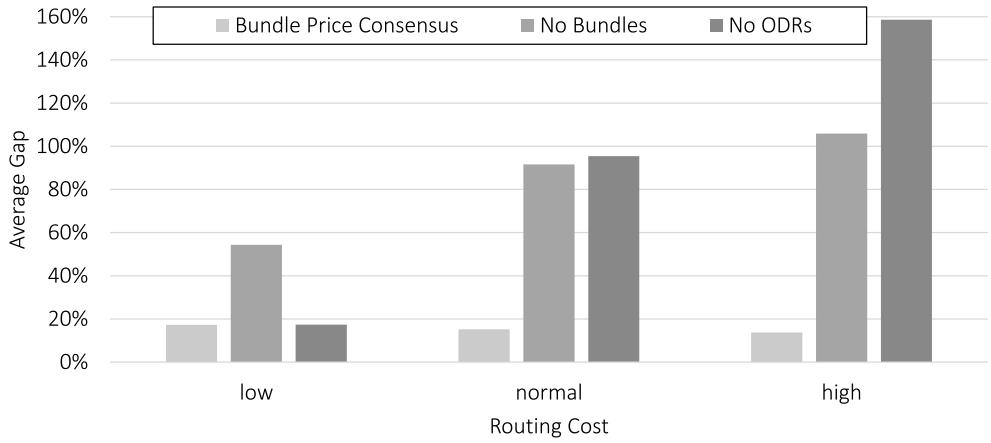


Fig. 10. Comparison of the perfect information gaps for the business concept-oriented benchmark policies for different routing cost.

the detour cost is smaller due to the triangular inequality. Second, the willingness to participate for ODs is higher when bundles are offered. The reason is the same, that the suppliers earn more by traveling relatively less per delivery. When analyzing the value of using ODs for delivery, we observe very different results with respect to the routing cost of the company fleet. In case routing cost are low, BPC (17.3% gap) and No ODs (17.4% gap) perform nearly the same. The value of using ODs is very small and companies may decide against the operational complexity of adding ODs to their delivery operations. Once the cost increases to normal, the benefit of BPC becomes significant (15.2% vs. 95.4% gap). Companies should consider adding ODs to their delivery concepts in case bundles are assigned in an anticipatory manner. Adding ODs but only assigning individual orders does not yield a larger benefit though (91.6% vs. 95.4% gap). In the case of high routing cost, ODs become particularly valuable, even assigning individual orders (105.9% vs. 158.6% gap), but especially when assigning bundles with BPC (13.7% vs. 158.6% gap).

6. Conclusion

In this work we have shown how the dynamic and anticipatory assignment of order bundles to occasional drivers can reduce delivery cost substantially. There are a variety of avenues for future research. In our problem, we introduced the concept of bundles to dynamic delivery with crowdsourced couriers. We focused on the case where all orders are known a priori and have to be picked up at the same location by in-store customers. Remaining unassigned orders are delivered by the company fleet at the end of the process. Future research may extend the problem in several dimensions. First, instead of a final routing in the end of the process, the company may decide to dynamically route delivery vehicles to ensure fast delivery or even time windows [40]. This increases complexity since now dynamic decisions are made about the dispatching of the vehicles and the unassigned orders to deliver. Such orders should be located close to each other and, at the same time, should not break up attractive bundles. Here, the idea of popularity might be used to filter for orders that are not part of any attractive bundle. Another extension may be that not only the arrivals of occasional drivers but also of orders are stochastic and dynamic [43]. In that case, bundles are created dynamically as well. While our threshold-concept might still apply in such settings, alternative measures for compensation prices and popularity must be developed to account for the uncertainty in the set of orders. Also, instead of considering a single pickup location (the store), delivery might be possible from multiple stores in the city or even well-located micro hubs nearby [44,45]. Here, decisions about the assignment of orders to stores must be made as well as potential transportation to the hubs. The proposed model and method may be connected with the work of [34] optimizing assignments of orders to in-store customers with both picking and routing in mind. Finally, the problem may be extended to a crowdsourced pickup and delivery problem as proposed in [23], or, even for crowdsourced passenger transportation [46].

We have proposed a threshold-based policy with scenario-based tuning that can be applied in real-time. The thresholds are time-dependent to account for decreasing assignment opportunities. Future research may extend this approach to adapt the thresholds dynamically,

e.g., for cases where more or less than expected orders remain in the system. This could be done via reinforcement learning as proposed in [47]. Alternatively, the thresholds may be adapted by carefully re-sampling a subset of scenarios. However, preliminary tests showed that besides an increase in runtime, the solution quality did not increase, but even declined, potentially due to the limited reliability of sampled scenarios and proxy cost in later stages of the process. Furthermore, it could be worthwhile combining the bundling and matching phase in each decision. Up to now it is not obvious how to adapt scenario policies such that an additional dimension (in this case the subadditivity of orders bundled to packages) can be integrated. Finally, future work should address scalability of the framework such that real-world sized instances can be solved. As several challenging combinatorial optimization problems have to be solved during the sequential decision process, the computational burden is very high. However, our study shows that the most time consuming part is the generation of bundles. Hence, in a first step, quick but smart heuristics for this part of the framework should be developed.

CRediT authorship contribution statement

Simona Mancini: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marlin W. Ulmer:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Margaretha Gansterer:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marlin Ulmer reports financial support was provided by German Research Foundation. Margaretha Gansterer reports financial support was provided by Austrian Science Fund. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Clearance policy

In the clearance policy the goal at each stage is to minimize the number of unserved customers and, in case of a tie on the first objective, to minimize the auction clearing cost. This goal is achieved by solving the following Integer Programming model.

$$\min \sum_{i \in I} Z_i + \epsilon \sum_{l \in L} C_l Y_l \tag{A.1}$$

$$Z_i + \sum_{l \in L | i \in \tau_l} Y_l \leq 1 \quad \forall i \in I \tag{A.2}$$

$$\sum_{l \in L | \tilde{o}_l = o} Y_l \leq 1 \quad \forall o \in O \tag{A.3}$$

$$Z_i \in \{0;1\} \quad \forall i \in I \tag{A.4}$$

$$Y_l \in \{0;1\} \quad \forall l \in L \tag{A.5}$$

The objective function is reported in (A.1). Constraints (A.2) impose that a request remains unserved if it is not inserted in a bundle selected in the auction, and that we cannot select more than one bundle containing the same request. Constraints (A.3) impose that at most one bundle is assigned to each driver. Finally, constraints (A.4) and (A.5) specify the variables domain. Sets I and L are updated at each stage and contain only the requests which have not be served yet and the bundles including only subsets of these requests, respectively.

Once the boundary condition is satisfied and there are still unserved requests a standard VRP is solved for serving them at the minimum cost possible with the owned fleet. This policy is myopic since, in the solution of each single stage, does not use any scenarios to take into account the potential impact on future stages of the decision made at the current stage. We compare the proposed approach against this policy, since we believe it is likely to be the most common policy used in practice.

Algorithm 2 Anticipatory Clearance Policy

$t \leftarrow 0$
$served \leftarrow 0$
$C \leftarrow 0$
$I^0 \leftarrow I$
$L^t \leftarrow L$
while $t \leq T$ **and** $served < |I|$ **do**
  $I^t \leftarrow I^{t-1}$
  $\hat{L}_t \leftarrow sample_{ODS}(t)$
  $L \leftarrow L + \hat{L}_t$
  $C^t \leftarrow solve_{clearance}(t)$
  $C \leftarrow C + C^t$
  $I^t \leftarrow I^t - \{served\ customers\}$
  $\hat{L}_t \leftarrow \hat{L}_t - \{bundles\ containing\ served\ customers\}$
  $served \leftarrow served + |\{served\ customers\}|$
end while
if $served < |I|$ **then**
  $C^{last} \leftarrow VRP(I^t)$
  $C \leftarrow C + C^{last}$
end if

A.2. Anticipatory clearance policy

This policy is an extension of the lookahead mechanisms. At each time slot, we samples future arrivals of ODs for the remaining time slots. The smaller the number of residual time slots, the smaller the number of ODs sampled. At the last time slot no future arrival is sampled. We solve the same optimization problem as in the *Clearance* policy (A.1)–(A.3), but we consider bids submitted by both real ODs and by the sampled ODs. All the bundles served by real ODs are assigned to them, while all those assigned to sampled ODs are kept in the system. A pseudocode of the policy is shown in Algorithm 2.

A.3. Further results

In this section, we present further results. We first investigate the impact of varying ϵ^B or ϵ^C. To this end, for the normal instance setting, we set one value to the best found value, 0.2 for both, and vary the other one between 0.0 and 0.5 in steps of 0.1. The results are shown

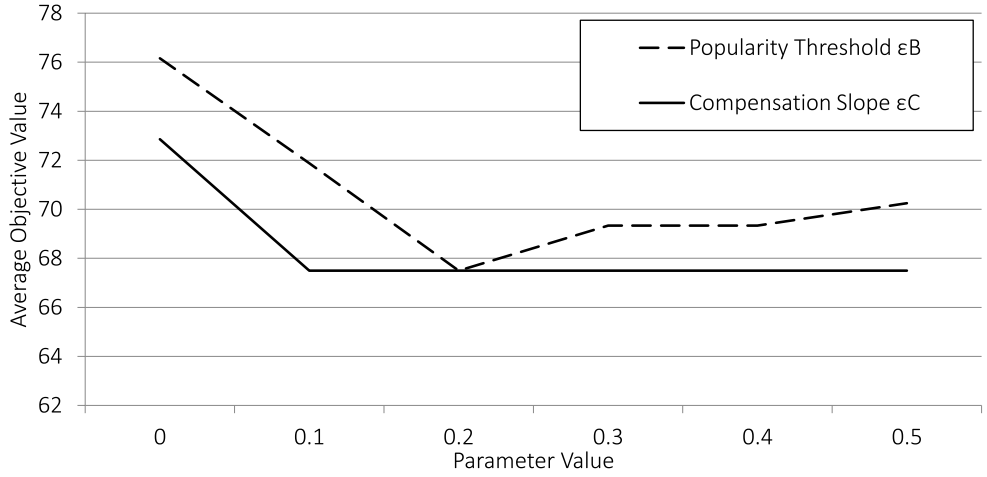


Fig. A1. Objective values for the normal instance setting and different values of either ϵ^B or ϵ^C.



Fig. A2. Observed detours for the normal instance setting and BPC.

in Fig. A1. The x-axis depicts the value for ϵ^B or ϵ^C, respectively. We observe that the best values are at 0.2. We further observe that setting the values to 0 yields inferior solution quality. For ϵ^B, we see that for larger values, cost increase again. Thus, the popularity threshold is rather sensitive to changes. In contrast, the slope value ϵ^C is constant if larger than 0.

Next, we analyze the distribution of detours for our BPC-policy and the normal instance setting. The results are shown in Fig. A2. About 50.8% of ODs travel a detour of less than one kilometer, 27.7% of ODs travel between one and two kilometers more, another 15.3% travel between 2 and 3 kilometers more, and only 6.2% more than 3 kilometers. None of the ODs has a detour of more than 4 kilometers. We observe that (at least with BPC), for the majority of ODs, the detour burden is quite manageable.

Data availability

Data will be made available on request.

References

[1] Boysen N, Emde S, Schwerdfeger S. Crowdshipping by employees of distribution centers: Optimization approaches for matching supply and demand. European J Oper Res 2022;296(2):539–56.

[2] Archetti C, Savelsbergh M, Speranza MG. The vehicle routing problem with occasional drivers. European J Oper Res 2016;254(2):472–80.

[3] Pourrahmani E, Jaller M. Crowdshipping in last mile deliveries: Operational challenges and research opportunities. Socio-Economic Plann Sci 2021;78:101063. http://dx.doi.org/10.1016/j.seps.2021.101063.

[4] Savelsbergh M, Ulmer MW. Challenges and opportunities in crowdsourced delivery planning and operations. 4OR 2022;20(1):1–21.

[5] Berger S, Bierwirth C. Solutions to the request reassignment problem in collaborative carrier networks. Transp Res Part E: Logist Transp Rev 2010;46:627–38.

[6] Gansterer M, Hartl RF, Sörensen K. Pushing frontiers in auction-based transport collaborations. Omega 2019. http://dx.doi.org/10.1016/j.omega.2019.01.011.

[7] Kaspi M, Raviv T, Ulmer MW. Directions for future research on urban mobility and city logistics. Networks 2022;79(3):253–63.

[8] Wang L, Xu M, Qin H. Joint optimization of parcel allocation and crowd routing for crowdsourced last-mile delivery. Transp Res B 2023;171:111–35.

[9] Buldeo Rai H, Verlinde S, Macharis C. Shipping outside the box. Environmental impact and stakeholder analysis of a crowd logistics platform in Belgium. J Clean Prod 2018;202:806–16. http://dx.doi.org/10.1016/j.jclepro.2018.08.210.

[10] Ermagun A, Stathopoulos A. Crowd-shipping delivery performance from bidding to delivering. Res Transp Bus Manag 2021;41:100614. http://dx.doi.org/10.1016/j.rtbm.2020.100614.

[11] Alnaggar A, Gzara F, Bookbinder JH. Crowdsourced delivery: A review of platforms and academic literature. Omega 2021;98:102139.

[12] Buldeo Rai H, Verlinde S, Macharis C. Who is interested in a crowdsourced last mile? A segmentation of attitudinal profiles. Travel Behav Soc 2021;22:22–31. http://dx.doi.org/10.1016/j.tbs.2020.08.004.

[13] Punel A, Ermagun A, Stathopoulos A. Studying determinants of crowd-shipping use. Travel Behav Soc 2018;12:30–40. http://dx.doi.org/10.1016/j.tbs.2018.03.005.

[14] Ta H, Esper TL, Hofer AR. Designing crowdsourced delivery systems: The effect of driver disclosure and ethnic similarity. J Oper Manage 2018;60:19–33. http://dx.doi.org/10.1016/j.jom.2018.06.001.
[15] Dahle L, Andersson H, Christiansen M, Speranza MG. The pickup and delivery problem with time windows and occasional drivers. Comput Oper Res 2019;109:122–33.
[16] Macrina G, Di Puglia Pugliese L, Guerriero F, Laganà D. The vehicle routing problem with occasional drivers and time windows. In: Sforza A, Sterle C, editors. Optimization and decision science: methodologies and applications. Cham: Springer International Publishing; 2017, p. 577–87.
[17] Di Puglia Pugliese L, Ferone D, Festa P, F. G, G. M. Solution approaches for the vehicle routing problem with occasional drivers and time windows. Optim Methods Softw 2022;37(4):1384–414.
[18] Ahamed T, Zou B, Farazi NP, Tulabandhula T. Deep reinforcement learning for crowdsourced urban delivery. Transp Res B 2021;152:227–57.
[19] Macrina G, Di Puglia Pugliese L, Guerriero F. Crowd-shipping with time windows and transshipment nodes. Comput Oper Res 2020;113:104806.
[20] Voigt S, Kuhn H. Crowdsourced logistics: The pickup and delivery problem with transshipments and occasional drivers. Networks 2022;79(3):403–26.
[21] Al Hla YA, Othman M, Saleh Y. Optimising an eco-friendly vehicle routing problem model using regular and occasional drivers integrated with driver behaviour control. J Clean Prod 2019;234:984–1001.
[22] Yu V, Aloina G, Jodiawan P, Gunawan A, Huang TC. The vehicle routing problem with simultaneous pickup and delivery and occasional drivers. Expert Syst Appl 2023;214:119118.
[23] Arslan A, Agatz N, Kroon A, Zuidwijk R. Crowdsourced delivery: A dynamic pickup and delivery problem with ad-hoc drivers. Transp Sci 2019;53(1):222–35.
[24] Dayarian I, Savelsbergh M. Crowdshipping and same-day delivery: Employing in-store customers to deliver online orders. Prod Oper Manage 2020;29(9):2153–74.
[25] Silva M, Pedroso JP, Viana A. Deep reinforcement learning for stochastic last-mile delivery with crowdshipping. EURO J Transp Logist 2023;12:100105.
[26] Tao Y, Zhuo H, Lai X. The pickup and delivery problem with multiple depots and dynamic occasional drivers in crowdshipping delivery. Comput Ind Eng 2023;182:109440.
[27] Gdowska K, Viana A, Pedroso JP. Stochastic last-mile delivery with crowdshipping. Transp Res Procedia 2018;30:90–100.
[28] Cheng SF, Chen C, Kandappu T, Lau HC, Misra A, Jaiman N, et al. Scalable urban mobile crowdsourcing: Handling uncertainty in worker movement. ACM Trans Intell Syst Technol 2017;9(3):1–24.
[29] Di Puglia Pugliese L, Ferone D, Macrina G, Festa P, Guerriero F. The crowd-shipping with penalty cost function and uncertain travel times. Omega 2023;115:102776.
[30] Torres F, Gendreau M, Rei W. Crowdshipping: An open VRP variant with stochastic destinations. Transp Res C 2022;140:103667.
[31] Ausseil R, Pazour JA, Ulmer MW. Supplier menus for dynamic matching in peer-to-peer transportation platforms. Transp Sci 2022;56(5):1304–26.
[32] Çınar AB, Dullaert W, Leitner M, Paradiso R, Waldherr S. The role of individual compensation and acceptance decisions in crowdsourced delivery. 2023, arXiv preprint arXiv:2305.01317.
[33] Alnaggar A, Gzara F, Bookbinder JH. Compensation guarantees in crowdsourced delivery: Impact on platform and driver welfare. Omega 2023;102965.
[34] Dayarian I, Pazour J. Crowdsourced order-fulfillment policies using in-store customers. Prod Oper Manage 2022;31(11):4075–94.
[35] Mancini S, Gansterer M. Bundle generation for last-mile delivery with occasional drivers. Omega 2022;108:102582.
[36] Triki C. Using combinatorial auctions for the procurement of occasional drivers in the freight transportation: A case-study. J Clean Prod 2021;304:127057.
[37] Shen CW, Hsu CC, Tseng KH. An auction-based multiagent simulation for the matching problem in dynamic vehicle routing problem with occasional drivers. J Adv Transp 2022;115:2999162.
[38] Fleckenstein D, Klein R, Steinhardt C. Recent advances in integrating demand management and vehicle routing: A methodological review. European J Oper Res 2023;306(2):499–518. http://dx.doi.org/10.1016/j.ejor.2022.04.032.
[39] Powell WB. Reinforcement learning and stochastic optimization. John Wiley & Sons, Ltd; 2022, http://dx.doi.org/10.1002/9781119815068.
[40] Baty L, Jungel K, Klein PS, Parmentier A, Schiffer M. Combinatorial optimization-enriched machine learning to solve the dynamic vehicle routing problem with time windows. Transp Sci 2024. http://dx.doi.org/10.1287/trsc.2023.0107.
[41] Soeffker N, Ulmer MW, Mattfeld DC. Stochastic dynamic vehicle routing in the light of prescriptive analytics: A review. European J Oper Res 2022;298(3):801–20.
[42] Gansterer M, Hartl RF, Vetschera R. The cost of incentive compatibility in auction-based mechanisms for carrier collaboration. Networks 2018;74(4):490–514.
[43] Liu S, Luo Z. On-demand delivery from stores: Dynamic dispatching and routing with random demand. Manuf Serv Oper Manage 2023;25(2):595–612.
[44] Mousavi K, Bodur M, Roorda MJ. Stochastic last-mile delivery with crowd-shipping and mobile depots. Transp Sci 2022;56(3):612–30. http://dx.doi.org/10.1287/trsc.2021.1088.
[45] Kızıl KU, Yıldız B. Public transport-based crowd-shipping with backup transfers. Transp Sci 2023;57(1):174–96. http://dx.doi.org/10.1287/trsc.2022.1157.
[46] Haferkamp J, Ulmer MW, Ehmke JF. Heatmap-based decision support for repositioning in ride-sharing systems. Transp Sci 2024;58(1):110–30. http://dx.doi.org/10.1287/trsc.2023.1202.
[47] Hildebrandt FD, Bode A, Ulmer MW, Mattfeld DC. 2023. Reinforcement learning variants for stochastic dynamic combinatorial optimization problems in transportation. FEMM Working Paper Series.